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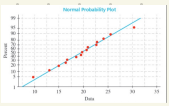
10.1

(13)



From the graphical duplay in number 13, the data seems to be approximately normal.
→ the points on the graph lies close to the line
there are no strong curve and bends

10.2

(20)

20. Cerebral Blood Flow Cerebral blood flow (CBF) in the brains of healthy people is normally distributed with a mean of 74. A random sample of 25 stroke patients resulted in an average CBF of $\bar{x} = 69.7$ with a standard deviation of $s = 16$. Test the hypothesis that the average CBF for stroke patients is different from that of healthy people using $\alpha = .05$.

H_0:
H_a

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}} = \frac{69.7 - 74}{16/\sqrt{25}} = \frac{-4.3}{3.2} = -1.343$$

$df = n-1 = 24$, $\alpha = 0.05$ (two-tailed)

$t_{crit} = \pm 2.064 \rightarrow -2.064 < -1.343 < 2.064$ ✗

At the 0.05 significance level, there is not enough evidence to conclude that the average CBF of stroke patients is different from healthy people

(30)

30. Bulimia An article in the *British Journal of Clinical Psychology* indicates that self-esteem was an important predictor of those who will benefit from treatment for bulimia.[4] The table gives the mean and standard deviation of self-esteem scores prior to treatment, at posttreatment, and during a follow-up:

	Pretreatment	Posttreatment	Follow-up
Sample Mean $\bar{x}$	20.3	26.6	27.7
Standard Deviation s	5.0	7.4	8.2
Sample Size n	21	21	20

a. Use a test of hypothesis to determine whether there is sufficient evidence to conclude that the true pretreatment mean is less than 25.
b. Construct a 95% confidence interval for the true posttreatment mean.
c. In Section 10.3, we will introduce small-sample techniques for making inferences about the difference between two population means. Without a formal statistical test, what might you conclude about the differences among the three sampled population means represented by the results in the table?

a) $H_0: \mu = 25$
$H_a: \mu < 25$

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}} = \frac{20.3 - 25}{5/\sqrt{21}} = \frac{-4.7}{1.09} = -4.31$$

$df = n-1 = 20$, $\alpha = 0.05$ (left-tailed)

$t_{crit} = -1.725 \rightarrow -4.31 < -1.725$ ✓

There is sufficient evidence to conclude that the true pretreatment mean self-esteem score is less than 25.

b) confidence interval $= \bar{x} \pm t_{\alpha/2}\left(\frac{s}{\sqrt{n}}\right)$ $t_{0.025}, df = 20$, $t_{crit} = 2.086$

$26.6 \pm t_{0.025/2}\left(\frac{7.4}{\sqrt{21}}\right)$

$26.6 \pm 2.086 \times 1.62 = 3.38$

26.6 ± 3.38

$(23.22, 29.98)$

We are 95% confident that the true post treatment mean self-esteem score lies between 23.22 and 29.98.

c) sample means — pre treatment = 20.3
posttreatment = 26.6
follow-up = 27.7

Based on the sample means, self-esteem appear to increase after treatment and keep being higher at follow-up. It can be concluded that treatment may have positive & lasting effect.

(35)

35. Sodium Chloride Measurements of water intake, obtained from a sample of 17 rats that had been injected with a sodium chloride solution, produced a mean and standard deviation of 31.0 and 6.2 cubic centimeters (cm^3), respectively. Suppose that the average water intake for noninjected rats observed over a comparable period of time is 22.0 cm^3.

a. Do the data indicate that injected rats drink more water than noninjected rats? Test at the 5% level of significance.
b. Find a 90% confidence interval for the mean water intake for injected rats.

a) $H_0: \mu = 22$
$H_a: \mu > 22$

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}} = \frac{31 - 22}{6.2/\sqrt{17}} = \frac{9}{1.5} = 6$$

$df = 16$, $\alpha = 0.05$

$t_{crit} = 1.746 \rightarrow 6 > 1.746$ Reject H_0.

There is significant evidence at the 5% level that injected rats drink more water than non-injected rats.

b) $\bar{x} \pm t_{\alpha/2} \frac{s}{\sqrt{n}}$ with 90% CI ($\alpha = 0.1$)

$31 \pm t_{0.05} \times \frac{6.2}{\sqrt{17}}$

$31 \pm 1.746 \times 1.5$

31 ± 2.62 $(28.38, 33.62)$

10.3

(7)

Two-Sample Tests *Use the information provided in Exercises 7–8. State the null and alternative hypotheses for testing for a significant difference in means, calculate the pooled estimate of σ^2, the associated degrees of freedom, and the observed value of the t statistic. What is the rejection region using $\alpha = .05$? What is the p-value for the test? What can you conclude from these data?*

7.

Population 1	12	3	8	5	
Population 2	14	7	7	9	6

hypotheses
$H_0: \mu_1 = \mu_2$
$H_a: \mu_1 \neq \mu_2$ (two-tailed)

sample means
$$\bar{x}_1 = \frac{12+3+8+5}{4} = 7$$
$$\bar{x}_2 = \frac{14+7+7+9+6}{5} = 8.6$$

sample variances
$$\sigma^2_1 = \frac{(12-7)^2 + (3-7)^2 + (8-7)^2 + (5-7)^2}{3} = \frac{25+16+1+4}{3} = 15.33$$
$$\sigma^2_2 = \frac{(14-8.6)^2 + (7-8.6)^2 + (7-8.6)^2 + (9-8.6)^2 + (6-8.6)^2}{4} = \frac{41.2}{4} = 10.3$$

pooled variances
$$\sigma^2_p = \frac{(n_1-1)\sigma^2_1 + (n_2-1)\sigma^2_2}{n_1+n_2-2} = \frac{3(15.33) + 4(10.3)}{4+5-2} = \frac{87.19}{7} = 12.4557 \approx 12.46$$
$$\sigma_p = \sqrt{12.46} = 3.529 \approx 3.53$$

degrees of freedom
$df = n_1 + n_2 - 2$
$= 4 + 5 - 2 = 7$

test statistic
$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sigma_p\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$
$$\sqrt{\frac{1}{4} + \frac{1}{5}} = \sqrt{0.25 + 0.20} = \sqrt{0.45} = 0.671$$
$$= \frac{7 - 8.6}{3.53 \times 0.671} = \frac{-1.6}{2.37} = -0.675 \approx -0.68$$
$t = -0.68$

rejection region
$df = 7$, $\alpha = 0.05$ (two tailed)
$t_{crit} = 2.365 \rightarrow t < 2.365$ or $t > 2.365$

p-value
$t = -0.68$, $df = 7$
$t = 0.911$ — one tail area $= 0.25$
$0.68 < 0.911$
approx. $P(T > 0.68) \approx 0.26$

two-tailed, then
$p = 2 \times 0.26$
$p = 0.52 > 0.05$ fail to reject H_0.
There is no significant difference between the two population

(1)

Population	1	2
Sample Size	16	13
Sample Mean	34.6	32.2
Sample Variance	4.8	5.9

hypotheses

$H_0: \mu_1 = \mu_2$

$H_a: \mu_1 \neq \mu_2$

pooled variance

$$\sigma^2_p = \frac{(n_1-1)\sigma_1^2 + (n_2-1)\sigma_2^2}{n_1+n_2-2}$$

$$= \frac{15(4.8)+12(5.9)}{16+13-2} = 5.2889 \approx 5.29$$

$$\sigma_p = \sqrt{5.29} = 2.30$$

degrees of freedom

$df = n_1+n_2-2$

$= 16+13-2 = 27$

test statistics

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sigma_p \sqrt{\frac{1}{n_1}+\frac{1}{n_2}}} \qquad \sqrt{\frac{1}{16}+\frac{1}{13}} = 0.3799$$

$$= \frac{34.6-32.2}{2.3 \times 0.379} = \frac{2.4}{0.86} = 2.79$$

rejection region

$df = 27$, $\alpha = 0.05$ (two-tailed)

$t_{crit} = 2.052 \rightarrow t < -2.052$ or $t > 2.052$

p-value

$t = 2.79$, $df = 27$

(look at t-table)

$2.79 > 2.771$ (one-tail probability < 0.005)

$p < 2 \times 0.005$

$p < 0.01$

$p \approx 0.009 < 0.05$ Reject H_0.

There is a significant difference between the two population means

(12)

12. Books or iPads? An experiment was conducted to compare the use of iPads versus regular textbooks in teaching algebra to two classes of middle school students.[8] To remove teacher-to-teacher variation, the same teacher taught both classes, and all teaching materials were provided by the same author and publisher. Suppose that after 1 month, 10 students were selected from each class and their scores on an algebra advancement test recorded. The summarized data follow.

	iPad	Textbook
Mean	86.4	79.7
Standard Deviation	8.95	10.7
Sample Size	10	10

a. Use the summary data to test for a significant difference in advancement scores for the two groups using $\alpha = .05$.

b. Find a 95% confidence interval for the difference in mean scores for the two groups.

c. In light of parts a and b, what can we say about using iPads versus traditional textbooks in teaching algebra at the middle school level?

a) $H_0: \mu_1 - \mu_2 = 0$

$H_a: \mu_1 - \mu_2 \neq 0$

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1}+\frac{s_2^2}{n_2}}} = \frac{86.4-79.7}{\sqrt{\frac{80.1025}{10}+\frac{114.49}{10}}} = \frac{6.7}{4.41} = 1.5189 \approx 1.52$$

calculate variance

$s_1^2 = 8.95^2 = 80.1025$

$s_2^2 = 10.7^2 = 114.49$

$df = 10+10-2 = 18$, $\alpha = 0.05$ (two tailed)

$t_{crit} = 2.101 \longrightarrow 1.52 < 2.101$ Fail to reject H_0

There is no sufficient evidence to conclude that there is a significant difference in advancement scores for the two groups.

b) 95% CI

$$(\bar{x}_1 - \bar{x}_2) \pm t_{\alpha/2} \times \sqrt{\frac{s_1^2}{n_1}+\frac{s_2^2}{n_2}}$$

$$(86.4-79.7) \pm t_{0.05/2} \times 4.41$$

$$6.7 \pm 2.101 \times 4.41$$

$$6.7 \pm 9.265$$

$$(-2.565, 15.965)$$

c) There is no statistically significant evidence at the 0.05 level that using iPads leads to different algebra test scores compared to traditional textbooks.

Eventhough ones using iPads had higher mean score, the difference is not statistically significant.